Social Network Analysis

Course Intro.

Dr. Chun-Hsiang Chan

Department of Geography,
National Taiwan Normal University

Photo credit: midjourney

Outline

- Lecturer introduction
- What is the social network?
- Why do you need to take this course?
- What will you learn from this course?
- Syllabus
- Grading policy

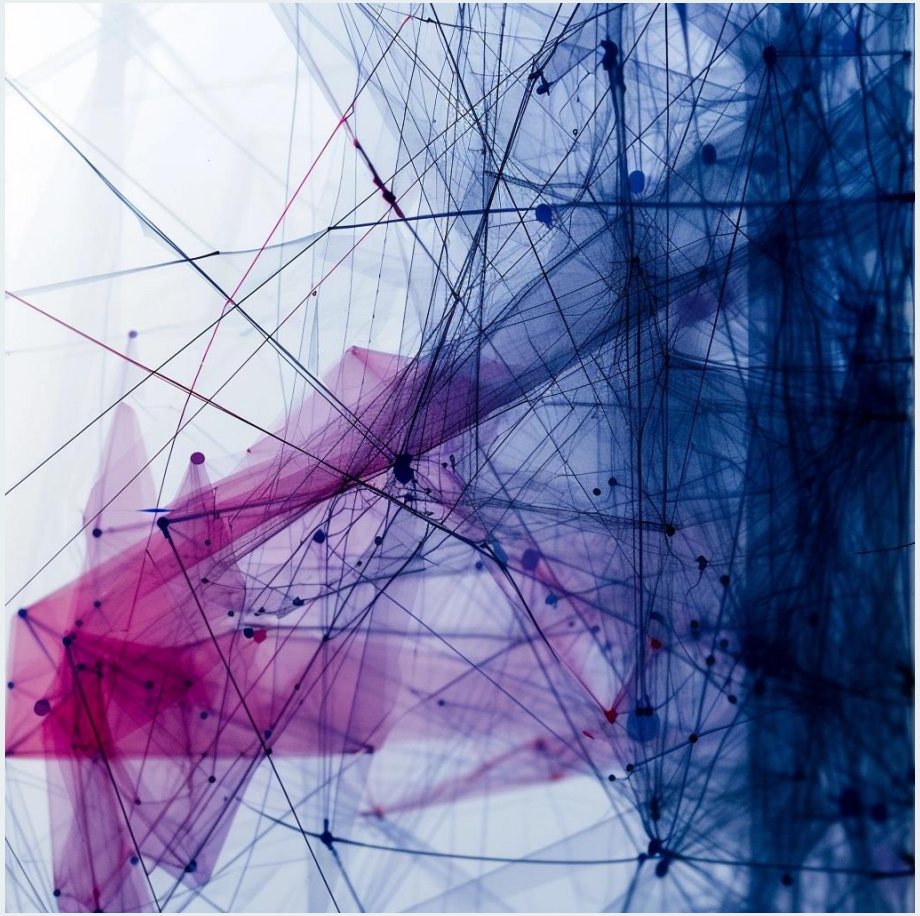


Photo credit: midjourney

Lecturer intro.

Present Position

- Assistant Professor | Department of Geography, National Taiwan Normal University

Education

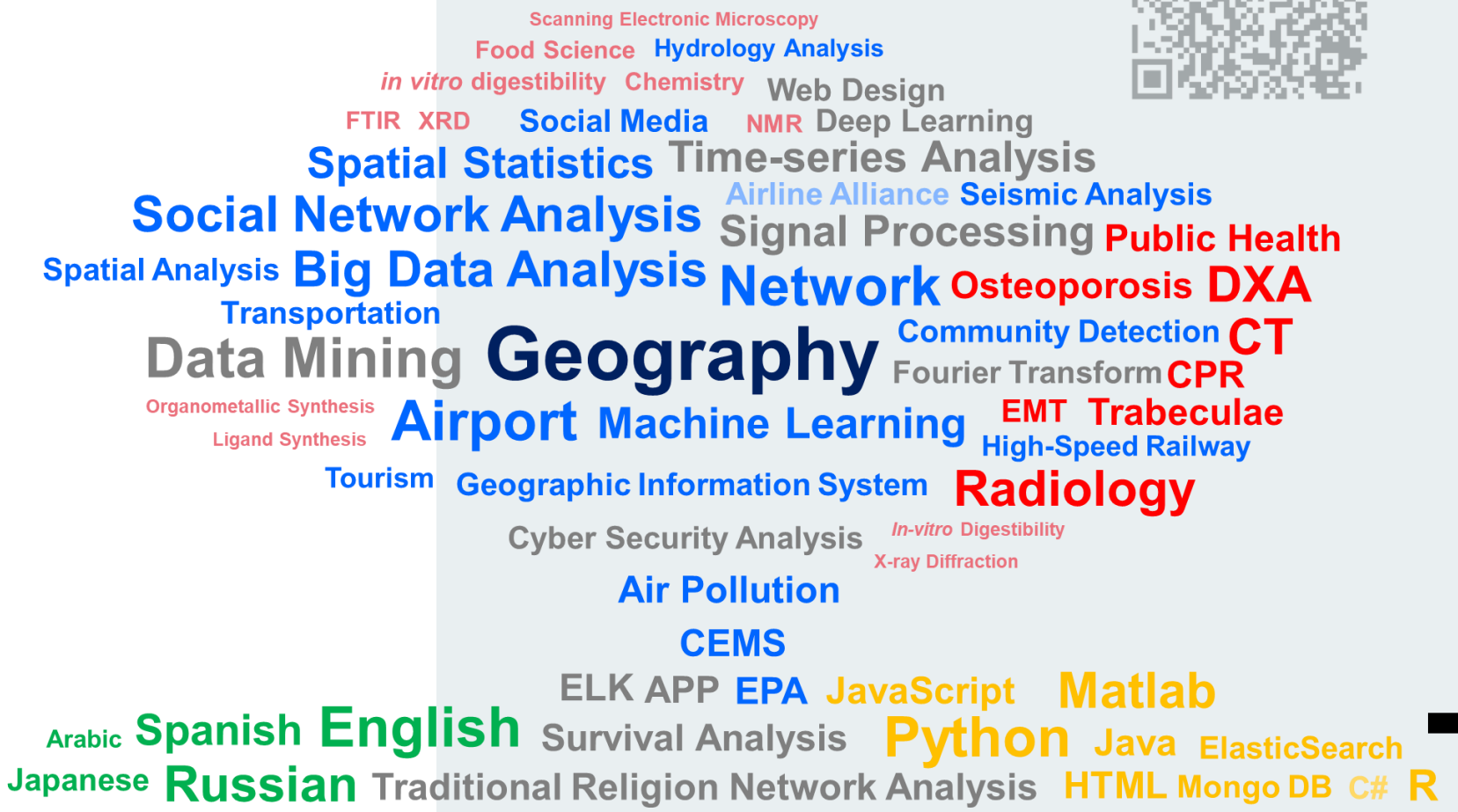
- Ph.D. | Department of Geography, National Taiwan University
- M.S. | Department of Geography, National Taiwan University
- M.S. | Department of Food Science, Nutrition, and Nutraceutical Biotechnology, Shih Chien University
- B.S.S. | Department of Social and Regional Development, National Taipei University of Education

Working Experience

- Assistant Professor | Undergraduate/ Master Program in Intelligent Computing and Big Data, Chung Yuan Christian University
- AI Consultant | Taiwan Cybersecurity Foundry Company
- Adjunct Data Scientist | Institute of Sociology, Academia Sinica
- Adjunct Assistant Professor | Department of Artificial Intelligence, Tamkang University
- Postdoctoral Research Fellow | Department of Radiology, School of Medicine, Taipei Medical University
- Postdoctoral Research Fellow | Department of Radiology, Taipei Municipal Wan Fang Hospital, Taipei Medical University
- Data Scientist Engineer | Cybersecurity Technology Institute, Institute for Information Industry
- Intern Research Assistant | National Science and Technology Center for Disaster Reduction

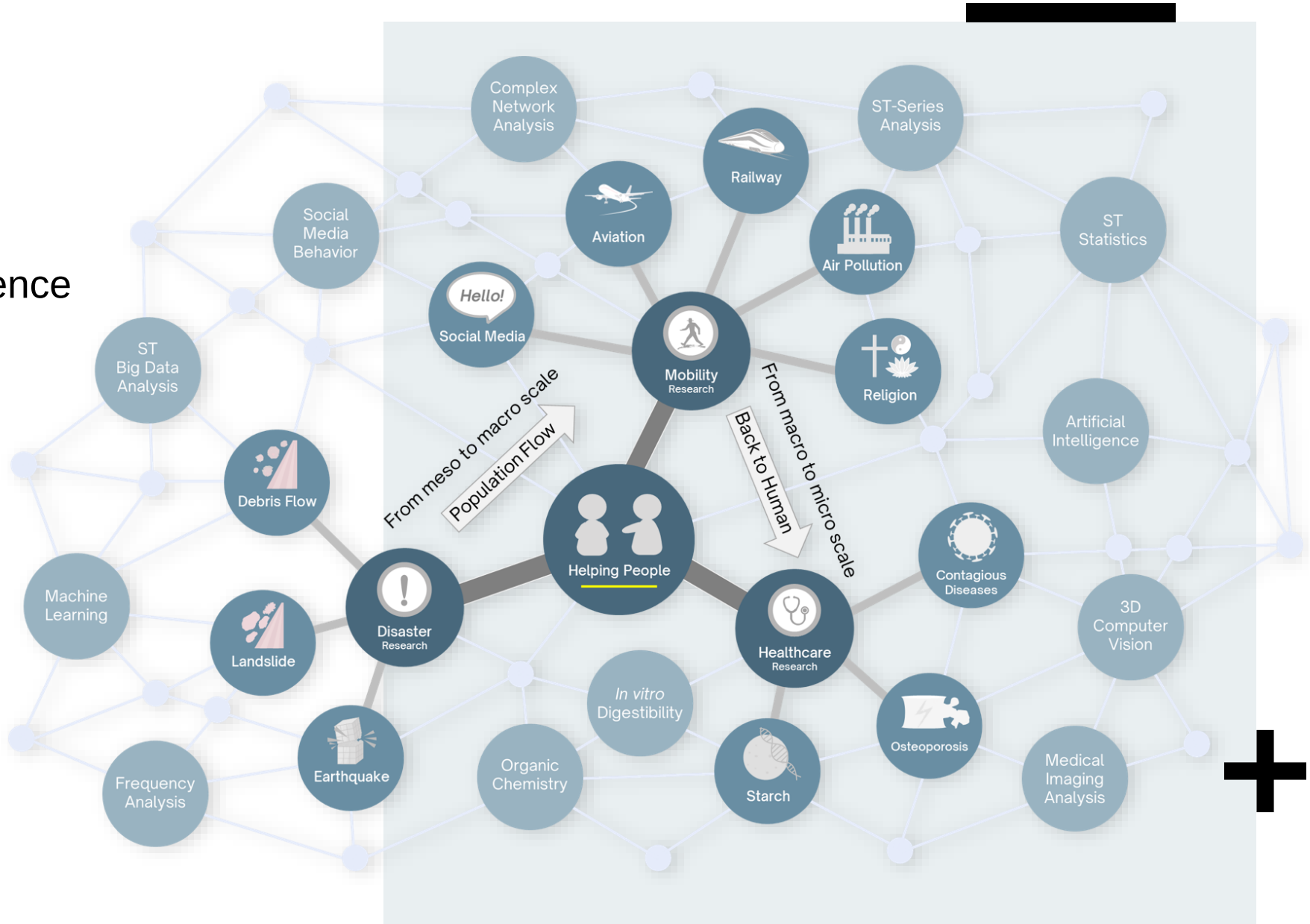
Skill tree

- Geography (GIS)
- Computer Science
- Radiology (bone & EM)
- Public Health
- Traditional Medicine
- Chemistry (basic & food)
- Cybersecurity
- Language (Russian)
- …



Research interests

- Spatiotemporal Data Science
- Artificial Intelligence (GeoAI & Explainable AI)
- Machine Learning
- Disaster Prevention
- Clinical Medicine (Ortho & EMS)
- Food Chemistry
- Cybersecurity



Research community

– My collaborators come from 29 academic institutions and companies across 6 countries.

What is the social networks?

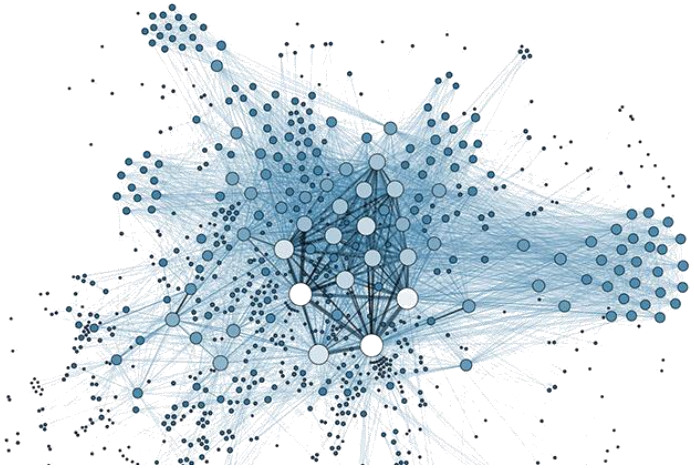
- **Complex networks are related to social networks.**
- A social network is a set of relations **[edge]** between subjects **[node]** (e.g., countries or people), and the relations could be anything (actual existence or abstract concept), such as trading volume and social media friendship.

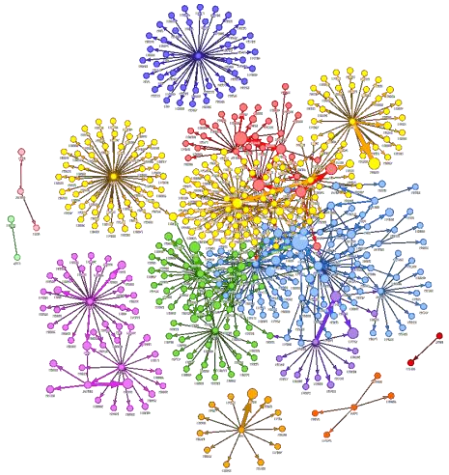
What is the social networks?

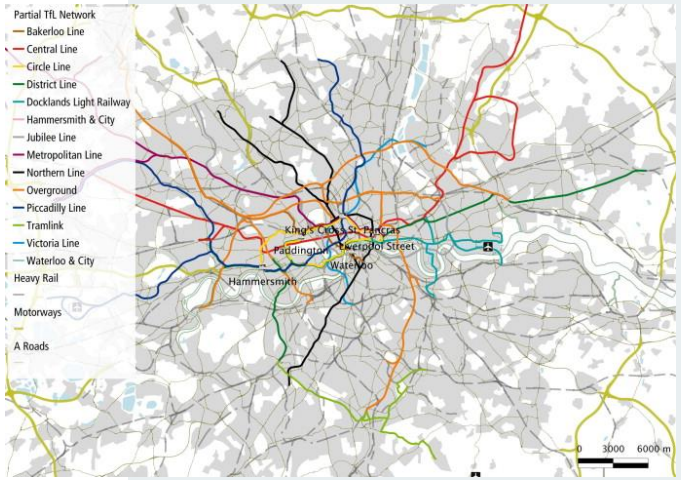
- We may leverage various quantitative parameters to characterize the network structure (node and edge).
- Social network analysis has been adopted in various non-sociological research, and the network structure has become much larger and more complicated; therefore, we call these types of complex networks.

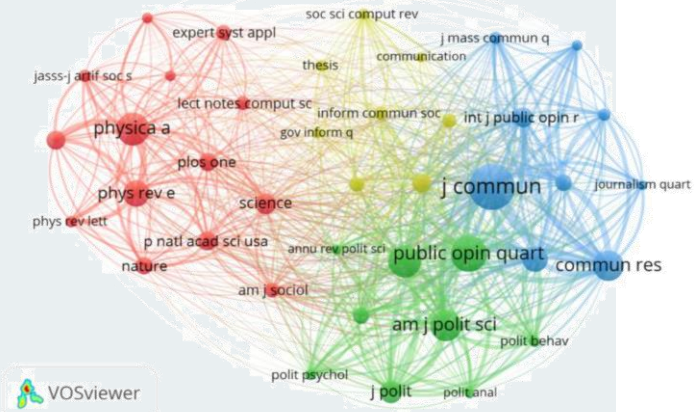
What is the social networks?

– Social networks have been utilized in a wide range of studies, including biological networks [1], religious networks [2], transportation networks [3], and citation networks [4].





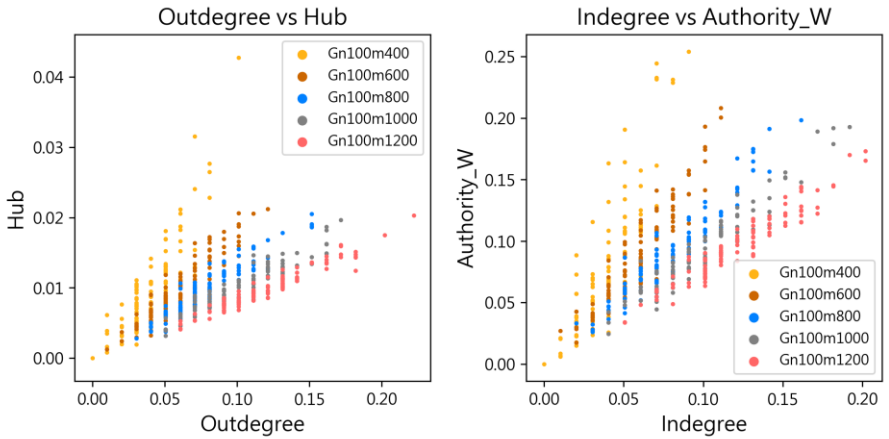


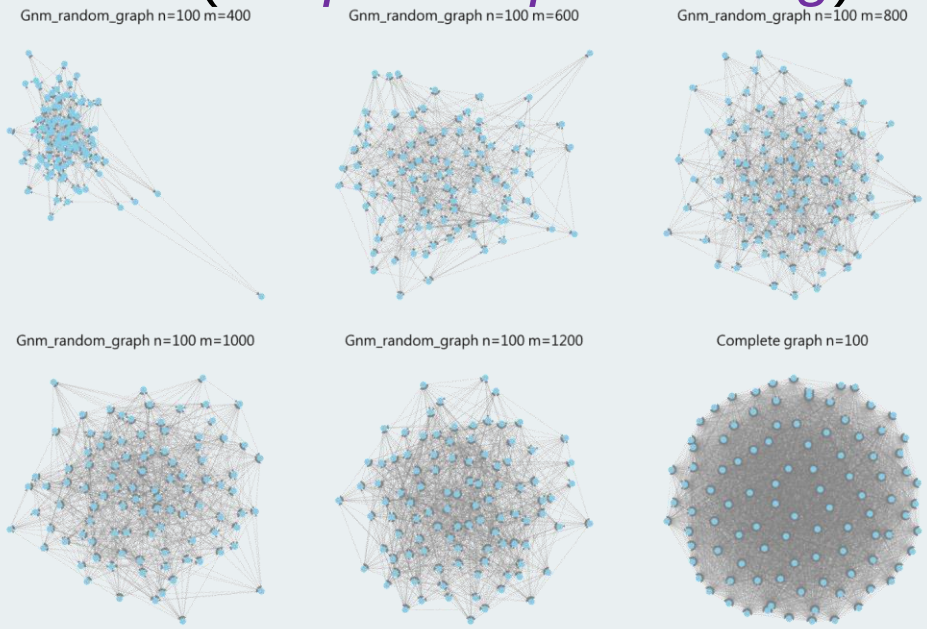


[1] https://serious-science.org/controlling-noisy-dynamics-in-biological-networks-to-fight-cancer-5376 (access on Jan. 21, 2023)
[2] Chun-Hsiang Chan, Wei-Hsian Chi. (2023) Soka Gakkai Feeling Network. Unpublished Work.
[3] Laura Ferrari, Michele Berlingerio, Francesco Calabrese, Jon Reades (2014) Improving the Accessibility of Urban Transportation Networks for People with Disabilities. Transportation Research Part C: Emerging Technologies. Vol 45, pp27-40.
[4] Yujie Qiang, Xuewen Tao, Xiaoqing Gou, Zhihui Lang, Hui Liu (2022) Towards a Bibliometric Mapping of Network Public Opinion Studies. Information. Vol. 13(1), 17; https://doi.org/10.3390/info13010017.

Why do you need to take this course?

– This course will teach the fundamental concepts of network formation and quantitative characterization for network structure, network visualization, community detection, and network autoregression. (*Graph Deep Learning*)





Syllabus

Week	Date	Content
1	Feb. 23	Course Introduction
2	Mar. 2	Introduction to Complex Network
3	Mar. 9	Data Collection and Network Formation *
4	Mar. 16	Ego Network *
5	Mar. 23	Centrality I *
6	Mar. 30	Centrality II
7	Apr. 6	Network Visualization
8	Apr. 13	Midterm Pitch
9	Apr. 20	Group

Week	Date	Content
10	Apr. 27	Network Positions
11	May 4	Two-mode Networks
12	May 11	Network Autoregression
13	May 18	Quadratic Assignment Procedure
14	May 25	Graph ML Intro.
15	Jun. 1	Final Project Presentation
16	Jun. 8	(Final Exams)

* Indicates online course

Grading policy

- All you have to do is study hard and feel free to ask questions when you do not understand.
- I believe that if you fulfill all the required items, you will pass this course.
- Do not worry about the grade! The most important thing is what you learn from this course.

Assignments	30 %	**Midterm Report**	20 %
Attendances	10 %	**Final Report**	40 %

Requirements

- In this course, we will **NOT** teach you how to design a program with Python.
- So, if you are **NOT** familiar with Python, please put **MORE EFFORT** into learning Python programming by any tutorials.
- Basically, … ***DON'T WORRY*** …

Textbooks

– Erwin Kreyszig (2018) Advance Engineering Mathematics 10e. Wiley. ISBN: 978-1-119-93416-5.
– Dmitry Zinoviev (2018) Complex Network Analysis in Python. Pragmatic Bookshelf. ISBN: 9781680502695.
– Krishna Raj P. M., Ankith Mohan, K. G. Srinivasa (2018) Practical Social Network Analysis in Python. Springer.

Assignments

– Small Mathematical Problems

– Coding Practice

– Paper Reading

$$C_i = \frac{\lambda_G(\nu)}{\tau_G(\nu)},$$

$$\tau_G(\nu) = \mathrm{C}(k_i, 2) = \frac{1}{2} k_i(k_i - 1)$$

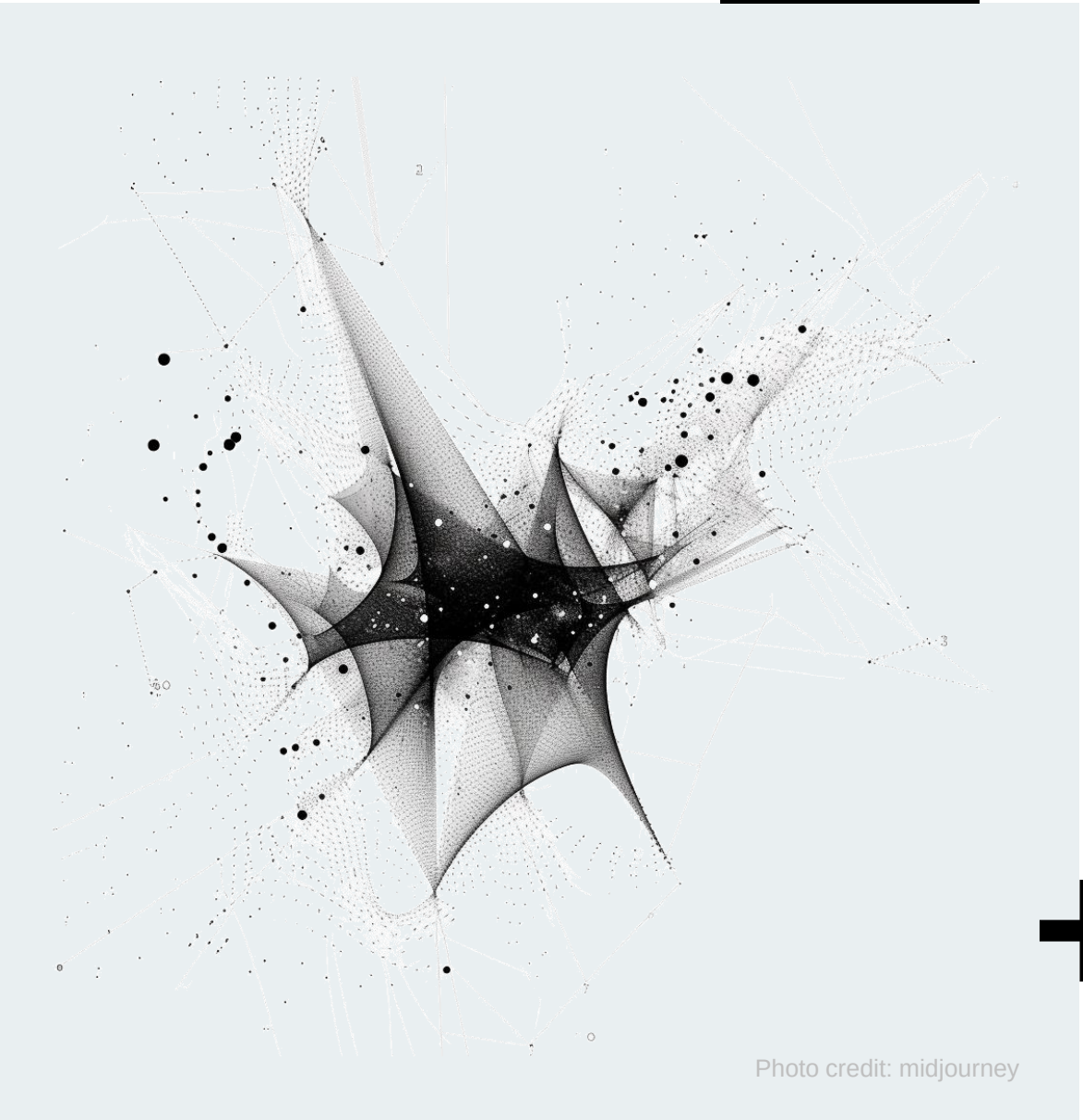


Photo credit: midjourney

The End

Thank you for your attention!

Email: chchan@ntnu.edu.tw
Website: toodou.github.io

Photo credit: midjourney